

Step-by-Step Problem Solving

Turn a vague business question into an auditable decision process.

TODAY'S PROBLEM

Why quick answers fail

- A short question hides the decision criteria.
- The model fills missing facts with plausible guesses.
- A polished recommendation can still be unusable.

BY THE END

You can build a decision prompt

- Separate facts, assumptions, and unknowns.
- Compare options with explicit criteria.
- Produce a recommendation that can be checked.

A decision is not a yes-or-no question

Before asking for a recommendation, define what must be true for it to be a good decision.

CASE

The supermarket decision

- A 12-person supermarket is considering two self-checkout kiosks.
- The owner wants shorter queues and lower labor pressure.
- The owner is worried about theft, downtime, and older shoppers.

HIDDEN QUESTIONS

What the first prompt usually omits

- What is the busiest hour and current queue time?
- What are the lease, support, training, and loss-prevention costs?
- What would count as a successful pilot after 90 days?

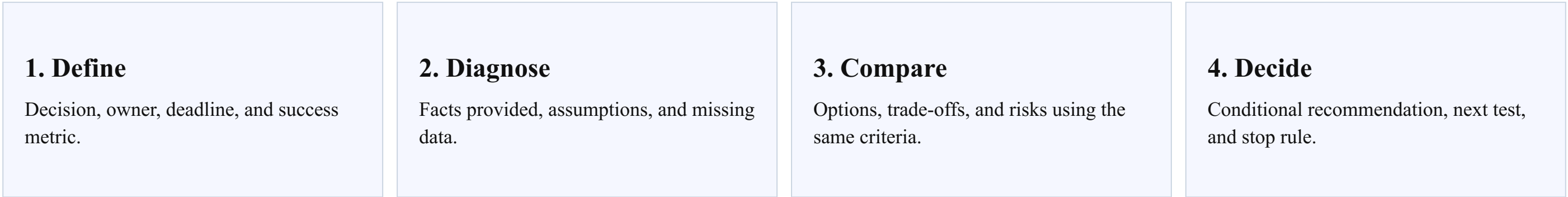
STUDENT RULE

Do not let the tool invent the brief

- List the decision, evidence, constraints, and success metric first.
- Mark missing information as unknown.
- Ask for a conditional recommendation when evidence is incomplete.

Use a four-stage decision frame

The framework makes the reasoning visible without treating the model output as proof.



Key idea: A reliable answer shows what it knows, what it assumes, and what evidence would change the recommendation.

Bad prompt: a question with no decision frame

Students should be able to identify exactly what is missing before they improve it.

PROMPT TO AVOID

Should our grocery store replace cashiers with self-checkout kiosks?

WHY IT FAILS

- No store facts, budget, customer profile, or time horizon.
- No definition of success or acceptable risk.
- Invites a generic opinion instead of a testable decision.
- The answer may sound confident while resting on invented assumptions.

Repair move: Add the decision context before asking for a conclusion.

Bad prompt: forced certainty from incomplete data

More detail is not enough if the prompt still requires an unsupported answer.

PROMPT TO AVOID

We have 12 employees. Tell us exactly how much profit two kiosks will add next year and guarantee that customers will accept them.

WHY IT FAILS

- Requests a precise forecast without cost, traffic, or loss data.
- Asks for a guarantee where uncertainty is central.
- Does not request sources, assumptions, or a pilot design.
- Encourages fabricated numbers instead of an evidence request.

Repair move: Replace false precision with ranges, assumptions, and a data-collection plan.

Good prompt: structure the diagnosis first

This prompt teaches the model to separate evidence from assumptions before it recommends action.

COPY AND ADAPT

You are helping a supermarket owner evaluate two self-checkout kiosks.

Known facts:

- 12 employees
- Goal: reduce peak-hour queues
- Concerns: theft, downtime, and older shoppers

Create a table with three sections:

1. Facts provided
2. Assumptions that need validation
3. Data we need before deciding

Do not estimate costs or benefits that are not in the facts. End with five questions for the owner.

WHY IT WORKS

- Names the role and a concrete decision.
- Separates facts from unverified assumptions.
- Prevents invented financial claims.
- Produces questions the student can take into field research.

Use it when: You are early in the problem and do not yet have reliable operational data.

Good prompt: compare options and design a pilot

Once the evidence is available, request a recommendation with conditions, not a promise.

COPY AND ADAPT

Act as a retail operations analyst. Use only the facts below.

[Paste verified queue, labor, kiosk lease, support, and loss data here]

Compare three options: keep current checkout, pilot two kiosks, or deploy kiosks to all lanes. Evaluate each option on: customer wait time, total operating cost, shrinkage risk, staff workload, and service access.

Then recommend one option. State:

- the assumptions that drive the recommendation
- a 90-day pilot metric and target
- one stop rule that would end the pilot

If evidence is missing, say "Information needed" rather than guessing.

WHY IT WORKS

- Uses a shared comparison criteria for every option.
- Requires a measurable pilot instead of a one-time leap.
- Makes uncertainty explicit.
- Creates a decision record another person can audit.

Exercise 1: Build an assumption audit

Work in pairs. Use the supermarket case and submit one completed audit table.

STEP 1 | 8 MIN

Write the decision brief

- State the decision owner, deadline, and intended outcome.
- List three known facts and three unknowns.

STEP 2 | 12 MIN

Run the diagnosis prompt

- Use Good Example 1 as a template.
- Do not add made-up data to fill gaps.

STEP 3 | 15 MIN

Audit the response

- Highlight each assumption.
- Classify it as testable, risky, or irrelevant.

DELIVERABLE | 15 MIN

Submit one page

- Decision brief.
- Assumption table.
- Five evidence questions for the owner.

Exercise 2: Pre-mortem and pilot redesign

Assume the kiosk pilot failed after 90 days. Design the next decision, not an excuse.

FAILURE SIGNAL

What happened?

- Queue times rose at peak hours.
- Staff bypassed the loss-prevention check.
- Customer complaints increased.

YOUR TASK

Run a pre-mortem

- Identify three plausible root causes.
- For each cause, name the evidence you would inspect.
- Separate controllable causes from external conditions.

DELIVERABLE

Create a restart plan

- Three corrective actions.
- One revised pilot metric for each action.
- One stop rule and one escalation owner.

A reusable checklist for business decisions

Use this before you accept any AI-generated recommendation.

1 Frame the decision • Who decides? By when? What does success look like?	2 Protect the evidence • Give verified facts. Label unknowns. Do not reward guessing.	3 Compare consistently • Apply the same criteria to every option.	4 Test before scaling • Use a pilot metric, target, and stop rule.
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